Youngung Jeong

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+82-55-214-3694 | https://youngung.github.io | https://github.com/youngung

Education

Hanyang University, BS, Department of Materials Science and Engineering	03/01/2001 – 02/29/2008
POSTECH, MS, Graduate Institute of Ferrous Technology (supervisor: F. Barlat)	03/01/2008 – 02/28/2010
POSTECH, PhD, Graduate Institute of Ferrous Technology (supervisor: F. Barlat)	03/01/2010 – 02/28/2014

Experiences

Assistant, associate, full professor, Changwon National University, ROK	03/01/2017 – Present
Faculty mentor guest, Los Alamos National Laboratory, NM, USA	10/07/2025 – 08/31/2026
Short term visitor, Los Alamos National Laboratory, NM, USA	07/11/2024 – 08/16/2024
Guest Scientist (offsite), Los Alamos National Laboratory, NM, USA	02/01/2022 – 02/1/2024
Postdoc, POSTECH, ROK	12/01/2016 – 02/28/2017
Research Scientist, Clemson university, SC, USA	03/01/2016 – 11/30/2016
Postdoc, NIST, MD, USA	03/01/2014 – 02/29/2016
Research Affiliate, Los Alamos National Laboratory, NM, USA	04/01/2012 – 09/30/2012
Guest Researcher, NIST, MD, USA	06/01/2011 – 12/31/2011

Selected recent publications

- Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations — 2025

International Journal of Plasticity 191
José Victoria-Hernández*, *Y. Jeong**, Dietmar Letzig

- Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel — 2024

International Journal of Plasticity 181
B. Jeon, S.-Y. Lee, J. Lee, *Y. Jeong**

- A crystal plasticity finite element analysis on the effect of prestrain on springback — 2023

International Journal of Mechanical Sciences, 237
M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, *Y. Jeong**,

- Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading — 2021

International Journal of Plasticity 147
*Y. Jeong**, B. Jeon, C. N. Tomé

Government-funded projects

Mid-career Researcher Program (PI), National Research Foundation of Korea	2023 - 2027
BK21, National Research Foundation of Korea	2020 - 2027
Engineering Research Center, National Research Foundation of Korea	2018 - 2024
General Research Program (PI), National Research Foundation of Korea	2020 - 2023
Virtual engineering platform project, Ministry of Trade, Industry and Energy	2018 - 2020
General Research Program (PI), National Research Foundation of Korea	2017 - 2020

Awards

Chungwoong Award (청웅상), The Korean Institute of Metals and Materials, 2026

Skills

Constitutive modeling: Phenomenological plasticity and crystal plasticity modeling

Languages: Korean, English

Computer skills: Linux, Git

Programming languages: Fortran, Python (NumPy, SciPy, and matplotlib), Matlab, shell scripts

Experimental Mechanics: uniaxial tension, shear, hydraulic bulge test, biaxial tests, digital image correlation (DIC)

Diffraction experiments: X-ray diffraction, Electron back-scattered diffraction (EBSD), Residual stress analysis

GitHub repositories

Elasto-visco-plastic self-consistent model (private repo) github.com/youngung/evpsc

- Extended visco-plastic self-consistent model to account for elasticity (ΔEVPSC and σEVPSC)
- Stand-alone calculation capabilities
- Various example shell and python scripts as well as Jupyter notebook files
- User material subroutine (UMAT) of Abaqus/standard solver
- Written primarily in Fortran with Python and shell scripts

VPSC8 and EVPSC (private repo) `github.com/Jeonbohye/VPSC8-dEVPSC`

- Combines VPSC8 and EVPSC models
- Numerical implementation of VPSC, ΔEVPSC, and σEVPSC constitutive model
- Three separate build commands for executables of VPSC8, ΔEVPSC, and σEVPSC.

In-house Python scripts for texture analysis `github.com/youngung/texture3`

- Can plot contoured pole figures and inverse pole figures from discrete orientations
- Written entirely in Python with open-sourced libraries including matplotlib, NumPy and SciPy
- Can generate RGB maps of EBSD (electron back-scattered diffraction) scans

Misc.

- Board member of Korean Society for Technology of Plasticity 2023-present
- Editorial board member of Korean Journal of Metals and Materials 2019-present

Full list of publications

1. Application of Δ EVPSC to interpret mechanical behavior of AZ31B sheet in warm temperature — submitted

   Kyoungwon Hwang, Bohye Jeon, Jaeseong Lee, Hyuk-Jong Bong, Jinwoo Lee, *Y. Jeong**

2. Elasto-visco-plastic model for deformation of dual phase steels — submitted

   Anirban Patra*, *Y. Jeong**, Carlos N. Tome

3. A cluster-based mean-field crystal plasticity modeling framework to simulate dynamic recrystallization — submitted

   *Y. Jeong**, Carlos N. Tome, Andrea Rovinelli, Laurent Capolungo

4. Calibrating self-consistent crystal plasticity models beyond stress-strain fitting: mechanism-activity features to discover solution families — submitted

   Jinyeong Yu, Jaeseong Lee, *Y. Jeong**, T. Lee*

5. Reverse engineering material behavior using Bayesian inference and finite element analysis on ring-pull test — submitted

   B. Jeon, C. N. Tome, P. M. Beck, B. Eftinik, A. Talapatra, *Y. Jeong**, L. Capolungo*

6. (1012) Twinning, Slip, and Texture in AZ31 Mg under Uniaxial Compression: A Comparative Study of Schmid Factor and Crystal Plasticity Modeling — accepted

   Metals and Materials International

   Jongbin Go*, *Y. Jeong**, Myeong-heom Park*, Sangwon Lee, Si Gao, Nobuhiro Tsuji

7. Leveraging Machine Learning and Crystal Plasticity for Efficient Calibration of Yld2004-18p Anisotropic Yield Function — 2026

   International Journal of Mechanical Sciences, 328, 112039

   A. S. Ebrahim, J. Kim, *Y. Jeong*, T. Park, H. Lim, B. L. Kinsey, J. Ha*

8. Experimental and numerical investigations on the microstructure and tensile deformation behavior in flash-butt welded joints of boron steel and the effects of a post heating condition — 2026

   Materials Characterization, 238, 116631

   Joonoh Moon*, Gyuyeol Bae*, *Y. Jeong*, Min-Ji Kwon, Jaeseong Lee

9. A Scientific Benchmark for Elasto-Plastic Constitutive Modeling - Part II: Blind Predictions, Calibration Strategies, and Benchmark Results — 2026

   International Journal of Material Forming, 19:17

   Lorenz Maier, Erik Walz, Michael Hofmann, Joana Rebelo Kornmeier, Rongfei Juan, Lian Junhe, *Youngung Jeong*, Bohye Jeon, Martin Milch, Alper Guner, Emad Maawad, Wolfram Volk, Christoph Hartmann

10. Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations — 2025

    International Journal of Plasticity, 191, 104380

    José Victoria-Hernández*, *Y. Jeong**, Dietmar Letzig

11. Modeling deformation, recovery, and recrystallization of tantalum using a higher order elasto-viscoplastic self-consistent model — 2025

    Journal of the Mechanics and Physics of Solids, 194, 105925

I. A. Riyad, B. Clausen, D. J. Savage, *Y. Jeong*, D. W. Brown, M. Knezevic*

12. A critical discussion of elasto-visco-plastic self-consistent (EVPSC) models 2024
Journal of Materials Research Technology, 33, 7596-7609
B. Jeon, *Y. Jeong**, C. N. Tomé

13. Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel 2024
International Journal of Plasticity, 181, 104098
Bohye Jeon, Shin-Yeong Lee, Jinwoo Lee, *Youngung Jeong**

14. Thermal ratcheting of uranium simulated with a thermo-elasto-visco-plastic self-consistent polycrystal model 2024
Journal of Nuclear Materials, 597, 155159
*Youngung Jeong**, Carlos N. Tomé

15. Crystal plasticity finite element simulations on extruded Mg-10Gd rod with texture gradient 2024
Journal of Magnesium and Alloys, 12 , 3409-3430
Jaeseong Lee, Dirk Steglich, *Youngung Jeong**

16. A comprehensive analysis of cermet design and thermal cyclic stability via elasto-viscoplastic crystal plasticity modeling 2024
International Journal of Plasticity, 179, 104032
Glenn R Peterson, *Youngung Jeong*, Carlos N Tomé, Michael D Sangid*

17. Temperature-dependent behavior of CP-Ti interpreted via self-consistent crystal plasticity simulation 2024
Materials Science and Engineering: A, 890, 145904
Bohye Jeon, Min-Su Lee, Tea-Sung Jun, *Youngung Jeong**

18. Finite element analysis using elasto-visco-plastic self-consistent polycrystal model for E-form Mg sheet subjected to bending 2023
Journal of Magnesium and Alloys, 11, 1393-1407
B. Jeon, M. S. Kim, S. H. Choi, *Y. Jeong**

19. A crystal plasticity finite element analysis on the effect of prestrain on springback 2023
International Journal of Mechanical Sciences, 237, 107796
M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, *Y. Jeong**

20. Reconstructing orientation data from the images of IPF maps and ODF sections extracted from the literature: A data-collection method for machine learning 2023
International Journal of Plasticity, 159, 103467
L. Kaushik, K.-S. Park, J.-G. Kim, J. Lee, *Y. Jeong*, S.-H. Choi*

21. Prediction and validation of stress triaxiality assisted by elasto-visco-plastic polycrystal model 2022
Korean Journal of Metals and Materials, 60, 607-618
Jinhwa Park, *Youngung Jeong**

22. In-situ neutron diffraction study of lattice deformation behaviour of commercially pure titanium at cryogenic temperature 2022

Scientific Reports, 12, 3719

M.-S. Lee, T. Kawasaki, T. Yamashita, S. Harjo, Y.-T. Hyun, *Y. Jeong*, T.-S. Jun

23. Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading 2021

International Journal of Plasticity, 147, 103110

*Y. Jeong**, B. Jeon, C. N. Tomé

24. An efficient elasto-visco-plastic self-consistent formulation: Application to steel subjected to loading path changes 2020

International Journal of Plasticity, 135, 102812

*Y. Jeong**, C. N. Tomé

25. Modelling-assisted description of anisotropic edge failure in magnesium sheet alloy under mixed-mode loading 2020

International Journal of Mechanical Sciences, 181, 105680

*Y. Jeong**, Dirk Steglich

26. Extension of the VPSC model to account for elasto-visco-plastic behavior using a perturbed viscoplastic approach 2019

Modelling and Simulation in Materials Science and Engineering, 27, 085013

*Y. Jeong**, C. N. Tomé

27. Superior tensile fracture strength of hot isostatically pressed TiC–steel metallic composite fabricated by a novel infiltration 2019

Materials Science and Engineering: A, 764, 138260

S. J. Park, *Y. Jeong*, C. W. Kim, J. H. Lee, S. C. Cho, S. B. Lee, S. K. Lee, D. H. Kim, H. U. Hong*

28. Enhancement in viscoplastic self-consistent FLD prediction model and its application for austenitic and ferritic stainless steels 2019

Metals and Materials International, 25, 1548–1563

*Y. Jeong**, T. Manninen

29. A crystal plasticity model for describing the anisotropic hardening behavior of steel sheets during strain-path changes 2018

International Journal of Plasticity,

H. Kim, F. Barlat, Y. Lee, S. Zaman, C. S. Lee, *Y. Jeong**

30. A comparative study between micro- and macro-mechanical constitutive models developed for complex loading scenarios 2017

International Journal of Plasticity, 93, 212-228

*Y. Jeong**, F. Barlat, C. N. Tomé, W. Wen

31. Uncertainty in flow stress measurements using X-ray diffraction for sheet metals subjected to large plastic deformations 2016

Journal of Applied Crystallography, 49, 1991-2004

*Y. Jeong**, T. Gnäupel-Herold, M. Iadicola, A. Creuziger

32. Texture-based forming limit prediction for Mg sheet alloys ZE10 and AZ31 2016
International Journal of Mechanical Sciences, 117, 102-114
D. Steglich, *Y. Jeong**

33. Forming limit prediction using a self-consistent crystal plasticity framework: a case study for BCC fiber textures 2016
Modelling and Simulation in Materials Science and Engineering, 24, 055002
*Y. Jeong**, M.-S. Pham, M. Iaidocola, A. Creuziger, T. Foecke

34. Multiaxial constitutive behavior of an interstitial-free steel: measurements through X-ray and digital image correlation 2016
Acta Materialia, 112, 84-93
*Y. Jeong**, M. Iaidocola, T. Gnäupel-Herold, A. Creuziger

35. Effect of martensitic phase transformation on the behavior of 304 austenitic stainless steel under tension 2016
Materials Science and Engineering: A, 649, 174-183
H. Wang*, *Y. Jeong**, B. Calusen, Y. LiU, R. J. McCabe, F. Barlat, C. N. Tomé

36. Evaluation of biaxial flow stress based on Elasto-Viscoplastic Self-Consistent analysis of X-ray Diffraction Measurements 2015
International Journal of Plasticity, 66, 103-118
Y. Jeong, T. Gnaupel-Herold, F. Barlat, M. Iadicola, A. Creuziger, M.-G. Lee*

37. Application of crystal plasticity to an austenitic stainless steel 2012
Modelling and Simulation in Materials Science and Engineering, 20, 024009
*Y. Jeong**, F. Barlat, M.-G. Lee

38. Biaxial Deformation Behavior of AZ31 Magnesium Alloy: Crystal-Plasticity-Based Prediction and Experimental Validation 2012
International Journal of Solids and Structure, 49, 3551-3561
D. Steglich*, *Y. Jeong*, M. O. Andar, T. Kuwabara